HW-2

Analysis of Electromyography Signals

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2.1 Preparing the Signals

(1) First Trial Visualization

The first trial in Run-1 of Subject-1 corresponds to a **Point** movement (Cue trigger = 200, Task trigger = 201). Figure 1 shows the raw time-domain signals from all four EMG sensors with trigger annotations. The trial follows the expected sequence: Trial Start (1000) → Fixation (768) → Cue (200) → Task Execution (201) → End (202).

The extensor channels (ExtProx, ExtDist) show visible amplitude modulation during task execution compared to rest, particularly in ExtProx where muscle activation for the pointing movement is clearly evident. The flexor channels (FlxProx, FlxDist) show more subtle changes. Notably, the raw signals contain significant DC offsets and low-frequency drifts, motivating the need for appropriate filtering.

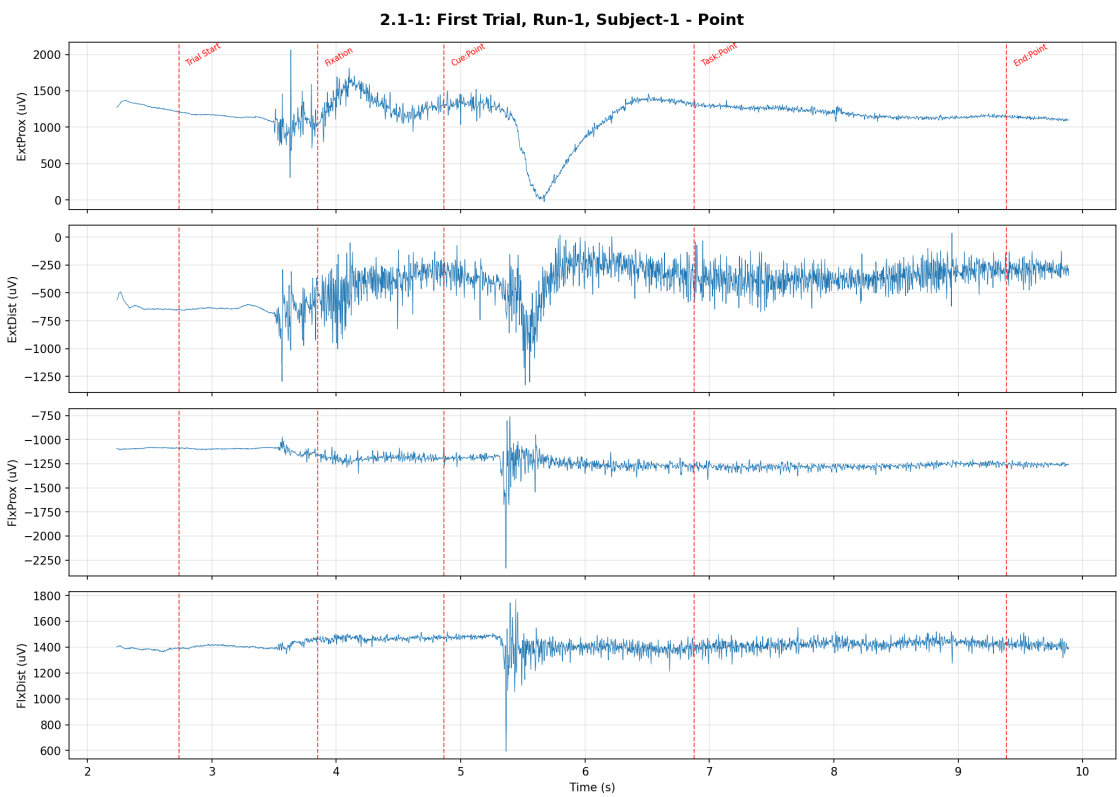

Figure 1: Raw EMG signals from all four sensors during the first trial (Point movement) in Run-1 of Subject-1. Red dashed lines indicate trigger positions: Trial Start, Fixation, Cue, Task Start, and Task End.

(2) PSD Analysis: Rest vs Task

Figure 2 shows the Welch PSD estimates for the rest and task periods of the distal flexor muscle (FlxDist) from the first trial. The task period exhibits higher power than the rest period across

the approximately **20–150 Hz range**, which is the characteristic frequency band for surface EMG activity. The greatest power difference between task and rest occurs in the 20–100 Hz band, consistent with the known spectral content of surface EMG signals where the bulk of muscle electrical activity resides.

A notable spectral peak is visible near **60 Hz** in both rest and task conditions, corresponding to powerline interference. This narrowband artefact must be addressed with a notch filter in addition to bandpass filtering.

With $f_s = 512$ Hz, the Nyquist frequency is 256 Hz, so the upper bound of the observable spectrum is inherently constrained. However, most surface EMG energy lies below 200 Hz, so this sampling rate is adequate for EMG analysis.

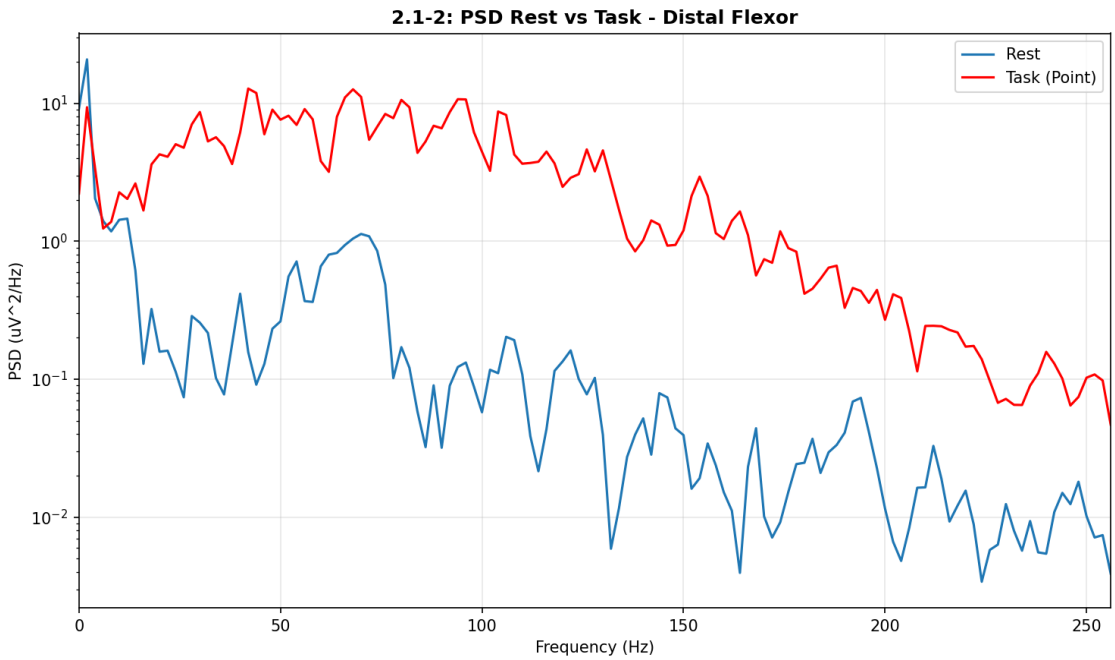

Figure 2: PSD comparison of Rest and Task (Point) periods for the distal flexor muscle (FlxDist) from the first trial. The task period shows elevated power in the 20–150 Hz range compared to rest. A 60 Hz powerline artefact is visible in both conditions.

(3) Filtering

Based on the PSD analysis, the following filtering pipeline was applied to all runs of both subjects:

1. **Bandpass filter**: 4th-order Butterworth, 20–200 Hz, zero-phase (`filtfilt`).
   - Below 20 Hz: motion artefacts and baseline drift dominate.
   - Above 200 Hz: approaching the Nyquist limit (256 Hz); contains mostly noise.
2. **Notch filter**: 60 Hz ($Q = 30$), zero-phase, to remove powerline interference.

The choice of 20–200 Hz is consistent with the surface EMG literature, where the dominant frequency content of the EMG signal typically falls within this band. Zero-phase filtering via `filtfilt` avoids phase distortion, which is important for preserving the temporal structure of the signal.

(4) Task Period Extraction

After filtering, task execution periods were extracted for all trials across all runs and both subjects using the task start (101/201/301) and task end (102/202/302) triggers. The extracted data was organised as:

- **Subject 1**: 180 trials (60 per class) $\times$ 4 sensors $\times$ 1282 samples
- **Subject 2**: 180 trials (60 per class) $\times$ 4 sensors $\times$ 1282 samples

Each task period is approximately 2.5 seconds long (1282 samples at 512 Hz), consistent with the experimental protocol.

2.2 Feature Extraction

(1) Features and Window Parameters

Two time-domain features were computed from the bandpass-filtered signals using a sliding window approach, following from HW-1:

Mean Absolute Value (MAV):

$$\text{MAV} = \frac{1}{N}\sum_{i=1}^{N}|x_i| \tag{1}$$

MAV captures the average amplitude of the EMG signal within a window, reflecting the overall level of muscle activation.

Variance (VAR):

$$\text{VAR} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2 \tag{2}$$

VAR measures the dispersion of the signal around its mean. As established in HW-1, VAR captures the variability of the signal and is sensitive to the modulation of neural firing patterns during muscle activation. VAR was selected as the second feature to maintain consistency with HW-1, where it was shown to yield higher SNR than MAV for detecting neural activity.

Window parameters. A window size of 250 ms (128 samples) with 50% overlap (step = 64 samples) was chosen. This yields approximately 19 feature windows per 2.5 s task period, providing a good balance between statistical stability (sufficient samples per window) and temporal resolution. For real-time prosthetic applications, 250 ms is within the acceptable latency range (< 300 ms) [1].

(2) Computed Features

Tables 1–4 summarise the computed MAV and VAR features (mean $\pm$ standard deviation across trials) for each movement class, each sensor, and each subject. Features were averaged across all sliding windows within each trial to produce one value per trial.

Table 1: MAV feature values (mean $\pm$ std, in μV) for Subject 1 across all runs.

Sensor	**Grasp**	**Pinch**	**Point**
ExtProx	54.74 ± 20.00	59.08 ± 22.58	38.38 ± 24.27
ExtDist	38.84 ± 12.19	32.39 ± 9.07	64.58 ± 12.56
FlxProx	15.28 ± 3.67	10.59 ± 4.13	22.86 ± 4.63
FlxDist	20.04 ± 3.43	20.55 ± 3.87	20.37 ± 2.68

From the tables, several patterns emerge. For Subject 1, ExtDist shows the clearest differentiation: Point has the highest MAV (64.58 μV) and VAR (7555 μV^2), well separated from Grasp and Pinch. FlxDist, on the other hand, shows nearly identical values across all three classes, suggesting

Table 2: VAR feature values (mean ± std, in μV^2) for Subject 1 across all runs.

Sensor	**Grasp**	**Pinch**	**Point**
ExtProx	6731 ± 3736	7973 ± 5335	4172 ± 4187
ExtDist	3041 ± 1617	2239 ± 1126	7555 ± 2994
FlxProx	434 ± 196	243 ± 180	964 ± 402
FlxDist	710 ± 239	775 ± 264	746 ± 188

Table 3: MAV feature values (mean ± std, in μV) for Subject 2 across all runs.

Sensor	**Grasp**	**Pinch**	**Point**
ExtProx	14.53 ± 3.61	17.91 ± 3.01	5.97 ± 1.76
ExtDist	10.50 ± 4.05	27.83 ± 4.63	29.93 ± 4.83
FlxProx	28.41 ± 8.52	23.02 ± 7.30	31.87 ± 11.32
FlxDist	18.28 ± 4.43	17.04 ± 4.18	18.84 ± 5.25

it provides little discriminative information. For Subject 2, ExtDist again differentiates well, with Pinch and Point having much higher values than Grasp. These patterns are consistent with the physiological roles of the extensor and flexor muscles during these hand movements.

(3) Feature Space Scatter Plots

Figures 3 and 4 visualise the 2D feature space (MAV vs VAR) for each sensor and subject, with each point representing one trial.

MAV and VAR are highly correlated, showing a strong positive nonlinear (approximately quadratic) relationship across all sensors. This is mathematically expected: for a zero-mean signal, $\text{VAR} \approx \text{MAV}^2$, so larger amplitude signals inherently have both higher MAV and higher VAR. This high correlation means that combining these two features provides limited additional discriminative information compared to using either alone.

Extensor channels (ExtProx, ExtDist) show the best class separation. In Subject 1's ExtProx, Grasp and Point trials occupy higher feature values while Pinch tends toward lower values. This is physiologically consistent: the extensor digitorum is heavily recruited during grasping and pointing movements.

Flexor channels (FlxProx, FlxDist) show more overlap between classes, especially FlxDist for Subject 1 where all three classes are heavily intermixed which is consistent with the near-identical FlxDist values in Table 1.

Subject 2 generally shows better class separation than Subject 1, likely reflecting individual differences in muscle activation patterns, electrode placement, or signal quality.

Table 4: VAR feature values (mean ± std, in μV^2) for Subject 2 across all runs.

Sensor	**Grasp**	**Pinch**	**Point**
ExtProx	394 ± 205	556 ± 166	85 ± 167
ExtDist	271 ± 308	1384 ± 347	1555 ± 446
FlxProx	1525 ± 843	977 ± 592	2051 ± 1337
FlxDist	600 ± 269	509 ± 225	657 ± 346

2.3 Average Patterns

(1–2) Average MAV Patterns: Proximal Flexor, Subject-1

The MAV envelope was computed following the procedure illustrated in Figure 4 of the assignment: bandpass filtering (20–200 Hz) → rectification (absolute value) → lowpass filtering at 5 Hz. The lowpass filter acts as a temporal smoother, producing a smooth amplitude envelope equivalent to computing MAV with a sliding window.

Figure 5 shows 5 random trials overlaid with the grand average for each movement class. Figure 6 compares the three average patterns directly.

Observations:

- The average patterns are relatively **flat** during the task period, without pronounced temporal dynamics. This suggests that muscle activation is sustained (tonic) rather than phasic during these isometric hand movements.
- **Point** produces the highest MAV (∼22–25 μV) in the proximal flexor, followed by **Grasp** (∼14–16 μV), and **Pinch** (∼9–11 μV). This amplitude ordering is physiologically meaningful: pointing with the index finger requires wrist stabilisation by the flexor carpi radialis, producing strong flexor activation; grasping involves moderate flexor activity for finger flexion; pinching is a fine motor task that primarily recruits smaller intrinsic hand muscles with less contribution from the forearm flexors.
- **Trial-to-trial variability** (grey lines) is substantial for all three classes, confirming the low SNR of individual EMG trials. The averaging process successfully reduces this variability: the average patterns (red lines) are much smoother and more distinct between classes, consistent with the noise-cancellation model in Eq. (1–2) of the assignment.

(3) Comparison with Feature Distributions

The amplitude ordering observed in the average patterns (Point > Grasp > Pinch) is **consistent with the feature distributions** in Section 2.2. In the FlxProx scatter plot of Subject 1, Point trials (blue) cluster at higher MAV/VAR values, Grasp (red) in the middle, and Pinch (green) at the lower end. The average patterns provide a more intuitive temporal view of the same phenomenon captured by the per-trial feature averages.

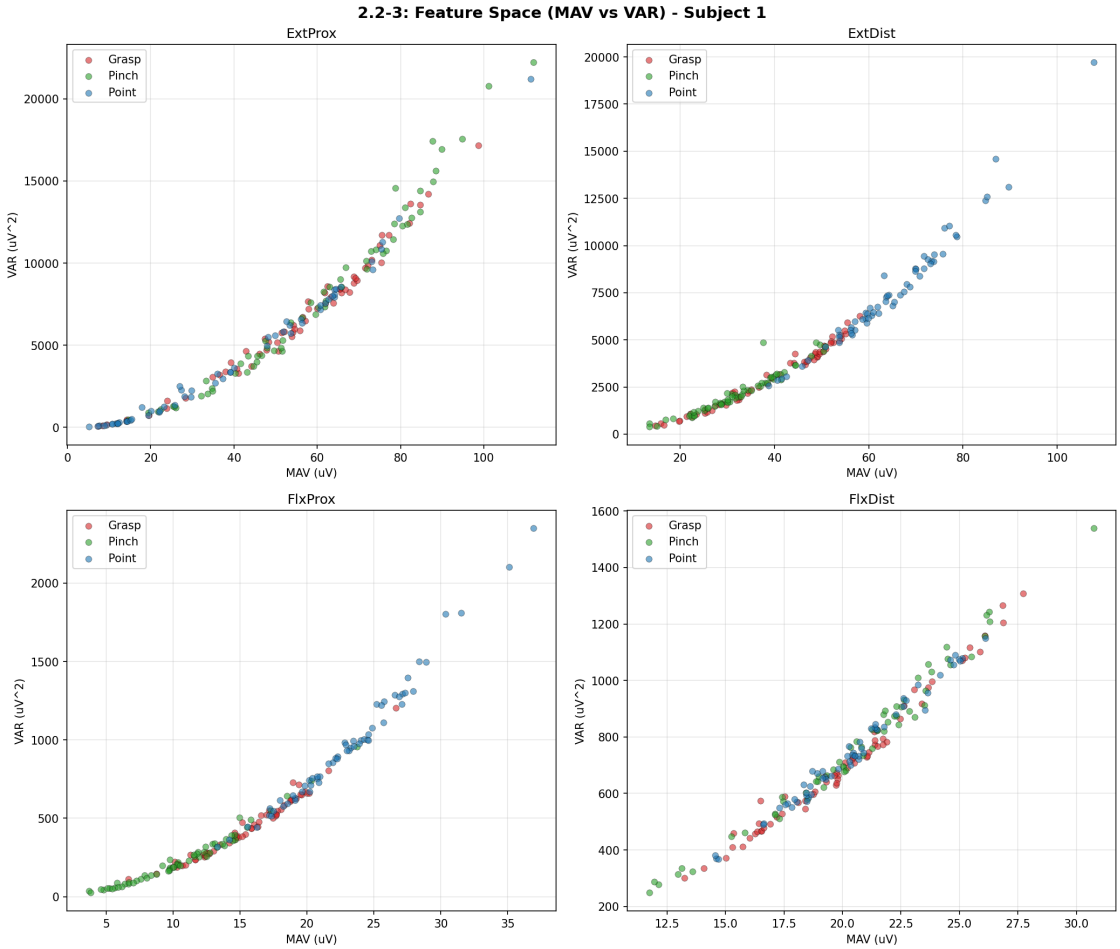

Figure 3: 2D feature space (MAV vs VAR) for all four sensors of Subject 1. Each point represents one trial, colour-coded by movement class. The extensor channels show better class separation than the flexor channels.

2.4 Classification

(1) Run-wise Cross-Validation

Both features (MAV + VAR) from all 4 sensors were concatenated to form an 8-dimensional feature vector per trial. Two classifiers were compared using leave-one-run-out cross-validation on Runs 1–5:

Table 5: Run-wise cross-validation accuracy (%) on Runs 1–5.

Classifier	Subject 1	Subject 2
LDA	78.0 ± 6.2	$\mathbf{95.3} \pm \mathbf{3.4}$
QDA	76.7 ± 6.3	92.0 ± 8.1

LDA outperforms QDA for both subjects, though the margin is small. With only 8 features

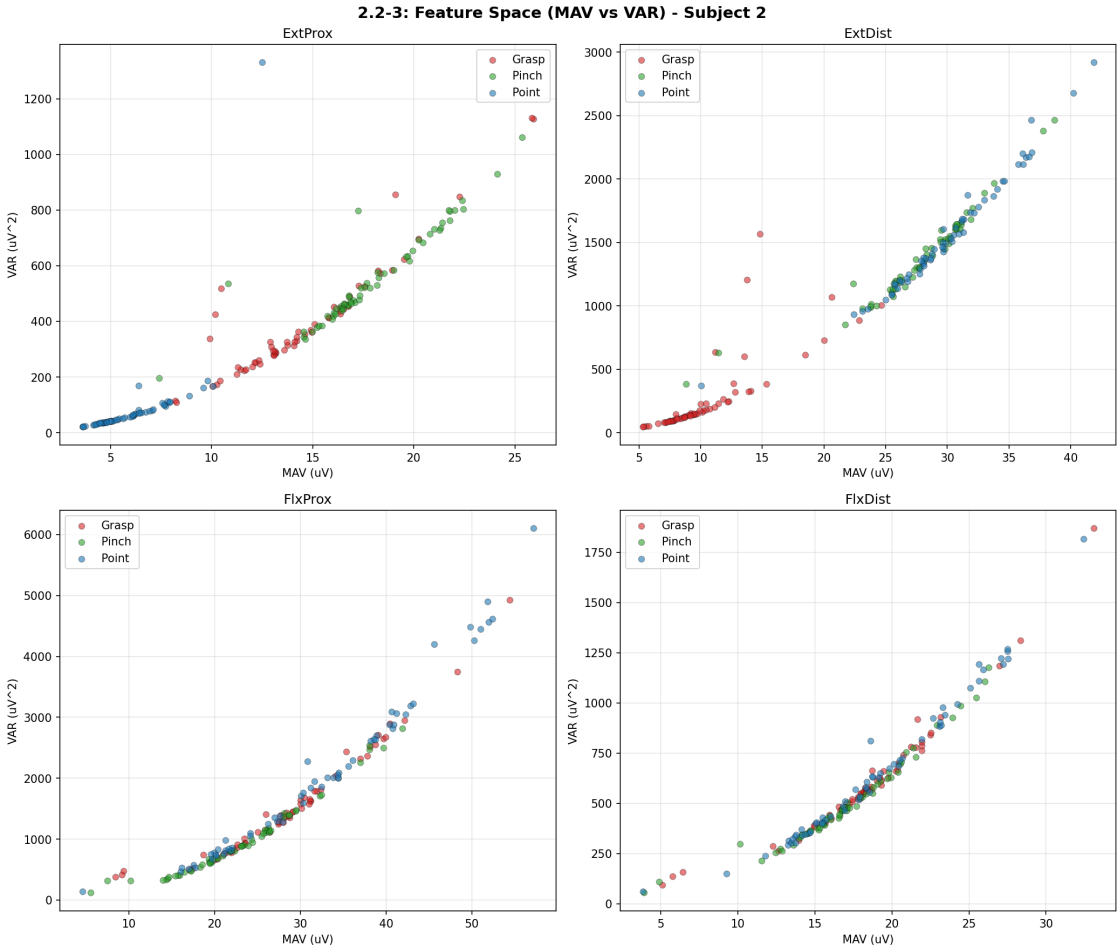

Figure 4: 2D feature space (MAV vs VAR) for all four sensors of Subject 2. Subject 2 shows generally better class separation compared to Subject 1.

and approximately 150 training samples (5 runs $\times$ 30 trials), QDA's additional parameters (separate covariance matrices per class) increase the risk of overfitting compared to LDA's shared covariance assumption. This is a classic bias-variance tradeoff: QDA has more flexibility but requires more data to estimate its parameters reliably.

Advantage of run-wise cross-validation. Run-wise CV ensures that training and test data come from temporally independent recording sessions. Within a single run, consecutive trials may share non-stationarities (e.g., muscle fatigue, electrode drift, subject adaptation). Random k-fold splitting would mix trials from the same run into both training and test sets, leading to **overly optimistic accuracy estimates** due to temporal autocorrelation, the same concern raised in HW-1 regarding fairness of cross-validation for time-series data.

(2) Test on Run-6

The best classifier (LDA) for each subject was trained on all of Runs 1–5 and tested on the held-out Run-6:

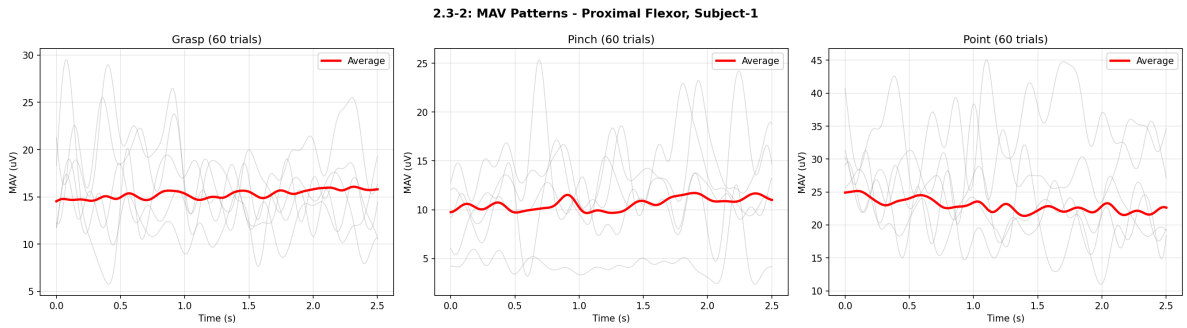

Figure 5: MAV envelope patterns for the proximal flexor muscle of Subject-1. Grey lines show 5 randomly selected individual trials; red line shows the grand average across all 60 trials per class.

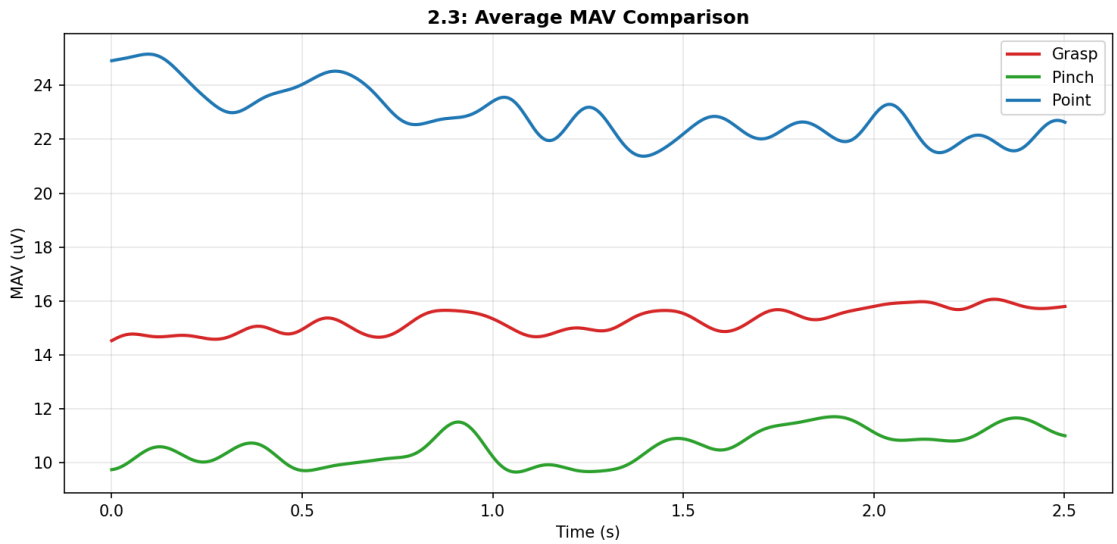

Figure 6: Comparison of average MAV patterns across all three movement classes for the proximal flexor of Subject-1. Point produces the highest activation, followed by Grasp and Pinch.

Subject 1's test accuracy (83.3%) is slightly higher than the CV average (78.0%), which can occur since CV averages over multiple folds while the test is a single evaluation on 30 trials. The confusion matrix (Figure 7) reveals that the main source of error is **Grasp being misclassified as Pinch** (3 out of 10 Grasp trials), which is consistent with these two movements having overlapping feature distributions, particularly in the flexor channels.

Subject 2 achieves perfect classification (100%), reflecting the generally better class separability observed throughout the analysis.

(3) Transfer Decoders

Each subject's LDA classifier (trained on Runs 1–5) was tested on the sixth run of the *other* subject:

Transfer accuracy drops near or below chance level (33.3% for 3 classes). The confusion matrices (Figure 8) confirm that the transferred classifiers fail to generalise across subjects. This severe

Table 6: Comparison of CV accuracy and Run-6 test accuracy.

	CV Accuracy	**Run-6 Test**
Subject 1	78.0%	83.3%
Subject 2	95.3%	100.0%

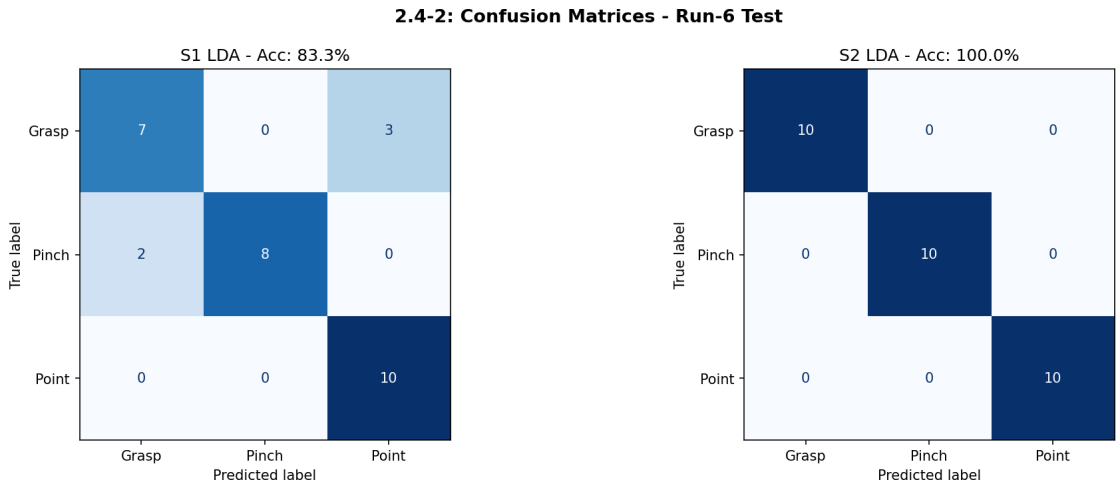

Figure 7: Confusion matrices for Run-6 test using LDA with MAV+VAR features. Subject 1 achieves 83.3% accuracy with Grasp–Pinch confusion as the primary error. Subject 2 achieves 100%.

degradation is expected and has several explanations:

1. **Inter-subject anatomical differences**: Variations in muscle size, subcutaneous fat thickness between the electrode and the muscle, and exact muscle geometry cause different EMG amplitude scales across subjects.

2. **Electrode placement variability**: Even small shifts in electrode location relative to the motor point of the muscle produce substantially different signal characteristics (amplitude, frequency content, crosstalk between muscles).

3. **Movement execution variability**: Each subject executes the same movement with slightly different motor strategies, recruiting different combinations of motor units.

4. **Feature distribution mismatch**: The LDA decision boundary learned for one subject's feature distribution is inappropriate for another subject's distribution, which may be shifted, scaled, or oriented differently in the 8D feature space.

Potential improvements for cross-subject generalisation include: feature normalisation (z-scoring per subject/session), domain adaptation techniques, or subject-specific calibration with a small number of labelled trials from the new subject.

Table 7: Transfer decoder accuracy on Run-6 of the other subject.

Transfer	**Accuracy**
S1 classifier → S2 Run-6	40.0%
S2 classifier → S1 Run-6	20.0%

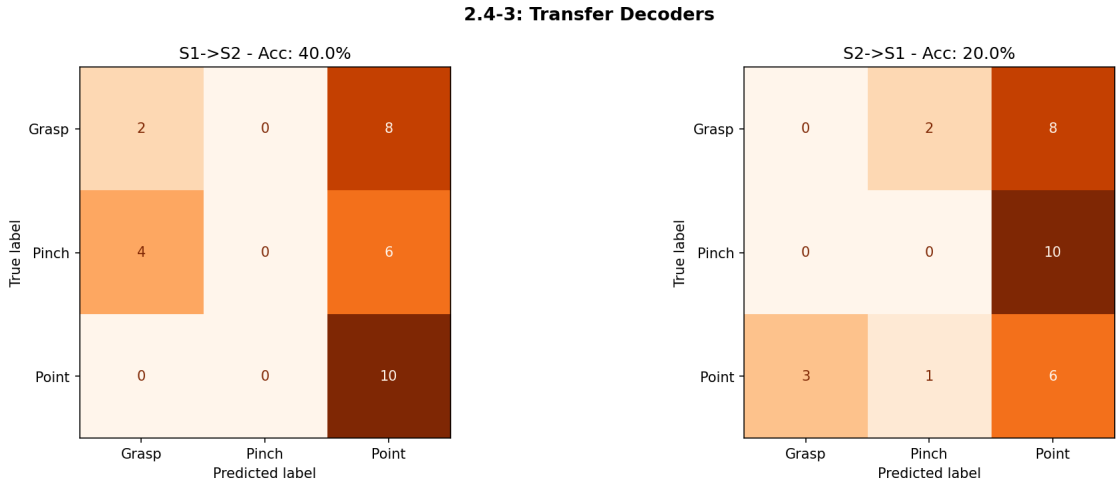

Figure 8: Confusion matrices for the transfer decoders. Left: Subject 1's classifier tested on Subject 2's Run-6 (40.0%). Right: Subject 2's classifier tested on Subject 1's Run-6 (20.0%). Both perform near or below chance level (33.3%).

References

[1] B. Hudgins, P. Parker, and R. N. Scott, "A new strategy for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 40, no. 1, pp. 82–94, 1993.

[2] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, 2003.